The Journal of Open Source Software

# cogsworth: A Gala of COSMIC proportions combining binary stellar evolution and galactic dynamics

**Tom Wagg** [1,2], **Katelyn Breivik** [3], **Mathieu Renzo** [2], **and Adrian M. Price-Whelan** [4]

**1** Department of Astronomy, University of Washington, Seattle, WA, 98195 **2** Center for Computational Astrophysics, Flatiron Institute, 162 Fifth Ave, New York, NY, 10010, USA **3** McWilliams Center for Cosmology and Astrophysics, Department of Physics, Carnegie Mellon University, Pittsburgh, PA 15213, USA **4** University of Arizona, Department of Astronomy & Steward Observatory, 933 N. Cherry Ave., Tucson, AZ 85721, USA

**DOI:** 10.xxxxxx/draft

**Software**

- Review
- Repository
- Archive

**Editor:** Ivelina Momcheva

**Reviewers:**

- @nicolagaspari
- @smandhai

**Submitted:** 06 September 2024
**Published:** unpublished

**License**
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

## Summary

We present `cogsworth`, an open-source Python tool for producing self-consistent population synthesis and galactic dynamics simulations. With `cogsworth` one can (1) sample a population of binaries and star formation history, (2) perform rapid (binary) stellar evolution, (3) integrate orbits through the galaxy and (4) inspect the full evolutionary history of each star or compact object, as well as their positions and kinematics. We include the functionality for post-processing hydrodynamical zoom-in simulations as a basis for galactic potentials and star formation histories to better account for initial spatial stellar clustering and more complex potentials. Alternatively, several analytic models are available for both the potential and star formation history. `cogsworth` can transform the intrinsic simulated population to an observed population through the joint application of dust maps, bolometric correction functions and survey selection functions.

## Statement of need

The majority of stars are born in binaries and multiple star systems (e.g., Duchêne & Kraus (2013); Moe & Di Stefano (2017); Offner et al. (2023)), a large subset of which will exchange mass at some point in their lifetime (e.g., Podsiadlowski et al. (1992); Sana et al. (2012); de Mink et al. (2014)). These massive stars play a critical role in the formation and evolution of galaxies as a result of their feedback (e.g., Dekel & Silk (1986); Hopkins et al. (2012); Nomoto et al. (2013); Somerville & Davé (2015); Naab & Ostriker (2017)). However, binary evolution remains uncertain, with many parameters such as common-envelope efficiency, mass transfer efficiency, angular momentum loss due to mass transfer and the mean magnitude of supernova natal kicks unconstrained across several orders of magnitude (e.g., Janka (2012); N. Ivanova et al. (2013); Katsuda et al. (2018); Natalia Ivanova et al. (2020); Röpke & De Marco (2023); Marchant & Bodensteiner (2023)).

Single massive stars are not expected to migrate far from their birth location before reaching core-collapse due to their short lifetimes ($\lesssim 50$,Myr, e.g., Zapartas et al. (2017)). However, binary stars may disrupt after an initial supernova event, ejecting the secondary star from the system with its orbital velocity (e.g., Blaauw (1961); Eldridge et al. (2011); Renzo et al. (2019)). Thus, close massive binaries that disrupt can lead to the displacement of secondary stars significantly farther from star-forming regions. The present-day positions and kinematics of massive stars and binary products are therefore strongly impacted by changes in binary physics that alter the separation prior to supernova. This means that comparing simulations of

positions and kinematics of stars and compact objects to observations will enable constraints on binary stellar evolution parameters.

The use of positions and kinematics as tracers of binary evolution has been considered in the past. Recent work has shown the importance of accounting for the galactic potential, which can change the velocity of kicked objects (e.g. Disberg et al. (2024)). It is also important to consider the inclination or timing of a supernova kick relative to the galactic orbit, since, for example, a kick out of the galactic plane at an object's highest galactic vertical position will have a strong effect on its final position. Failing to consider impacts from both a galactic potential and kicks (i.e. velocity impulses) will lead to misleading conclusions regarding the final spatial distributions of the population. Some studies have considered using the Galactic potential at the present-day position of objects to place a lower limit on the peculiar velocity at birth and constrain supernova kicks (Atri et al., 2019; Repetto et al., 2012, 2017; Repetto & Nelemans, 2015), but the accuracy of this method is debated (Mandel, 2016). Other work has considered the impact of the Galactic potential for individual special cases, rather than at a population level. For example, Evans et al. (2020) considered the orbits of hyper-runaway candidates evolving through the Milky Way potential, whilst Neuhäuser et al. (2020) developed software for tracing the motion of stars to investigate the recent nearby supernovae that ejected $\zeta$ Ophiuchi.

There is a clear need for an open-source tool that provides the theoretical infrastructure for making predictions for the positions and kinematics of massive stars and compact objects, placing these systems in the context of their host galaxy and its gravitational potential. `cogsworth` fulfils this need, providing a framework for self-consistent population synthesis and galactic dynamics simulations. The code is applicable to a wide range of binary products, both common and rare, from walkaway and runaway stars to X-ray binaries, as well as gravitational-wave and gamma-ray burst progenitors.

# Acknowledgements

We gratefully acknowledge many fruitful discussions with Julianne Dalcanton and Eric Bellm that resulted in several helpful suggestions. TW acknowledges valuable conversations with Matt Orr and Chris Hayward regarding the FIRE simulations, and with Alyson Brooks and Akaxia Cruz regarding the ChaNGa simulations. TW thanks the Simons Foundation, Flatiron Institute and Center for Computational Astrophysics for running the pre-doctoral program during which much of this work was completed. The Flatiron Institute is supported by the Simons Foundation. TW and KB acknowledge support from NASA ATP grant 80NSSC24K0768.

# References

Atri, P., Miller-Jones, J. C. A., Bahramian, A., Plotkin, R. M., Jonker, P. G., Nelemans, G., Maccarone, T. J., Sivakoff, G. R., Deller, A. T., Chaty, S., Torres, M. A. P., Horiuchi, S., McCallum, J., Natusch, T., Phillips, C. J., Stevens, J., & Weston, S. (2019). Potential kick velocity distribution of black hole X-ray binaries and implications for natal kicks. *489*(3), 3116–3134. https://doi.org/10.1093/mnras/stz2335

Blaauw, A. (1961). On the origin of the O- and B-type stars with high velocities (the "run-away" stars), and some related problems. *15*, 265.

de Mink, S. E., Sana, H., Langer, N., Izzard, R. G., & Schneider, F. R. N. (2014). The Incidence of Stellar Mergers and Mass Gainers among Massive Stars. *782*(1), 7. https://doi.org/10.1088/0004-637X/782/1/7

Dekel, A., & Silk, J. (1986). The Origin of Dwarf Galaxies, Cold Dark Matter, and Biased Galaxy Formation. *303*, 39. https://doi.org/10.1086/164050

DRAFT

Disberg, P., Gaspari, N., & Levan, A. J. (2024). Deceleration of kicked objects due to the Galactic potential. *687*, A272. https://doi.org/10.1051/0004-6361/202449996

Duchêne, G., & Kraus, A. (2013). Stellar Multiplicity. *51*(1), 269–310. https://doi.org/10.1146/annurev-astro-081710-102602

Eldridge, J. J., Langer, N., & Tout, C. A. (2011). Runaway stars as progenitors of supernovae and gamma-ray bursts. *414*(4), 3501–3520. https://doi.org/10.1111/j.1365-2966.2011.18650.x

Evans, F. A., Renzo, M., & Rossi, E. M. (2020). Core-collapse supernovae in binaries as the origin of galactic hyper-runaway stars. *497*(4), 5344–5363. https://doi.org/10.1093/mnras/staa2334

Hopkins, P. F., Quataert, E., & Murray, N. (2012). Stellar feedback in galaxies and the origin of galaxy-scale winds. *421*(4), 3522–3537. https://doi.org/10.1111/j.1365-2966.2012.20593.x

Ivanova, N., Justham, S., Chen, X., De Marco, O., Fryer, C. L., Gaburov, E., Ge, H., Glebbeek, E., Han, Z., Li, X.-D., Lu, G., Marsh, T., Podsiadlowski, P., Potter, A., Soker, N., Taam, R., Tauris, T. M., van den Heuvel, E. P. J., & Webbink, R. F. (2013). Common envelope evolution: where we stand and how we can move forward. *21*, 59. https://doi.org/10.1007/s00159-013-0059-2

Ivanova, Natalia, Justham, S., & Ricker, P. (2020). *Common Envelope Evolution*. https://doi.org/10.1088/2514-3433/abb6f0

Janka, H.-T. (2012). Explosion Mechanisms of Core-Collapse Supernovae. *Annual Review of Nuclear and Particle Science*, *62*(1), 407–451. https://doi.org/10.1146/annurev-nucl-102711-094901

Katsuda, S., Morii, M., Janka, H.-T., Wongwathanarat, A., Nakamura, K., Kotake, K., Mori, K., Müller, E., Takiwaki, T., Tanaka, M., Tominaga, N., & Tsunemi, H. (2018). Intermediate-mass Elements in Young Supernova Remnants Reveal Neutron Star Kicks by Asymmetric Explosions. *856*(1), 18. https://doi.org/10.3847/1538-4357/aab092

Mandel, I. (2016). Estimates of black hole natal kick velocities from observations of low-mass X-ray binaries. *456*(1), 578–581. https://doi.org/10.1093/mnras/stv2733

Marchant, P., & Bodensteiner, J. (2023). The Evolution of Massive Binary Stars. *arXiv e-Prints*, arXiv:2311.01865. https://doi.org/10.48550/arXiv.2311.01865

Moe, M., & Di Stefano, R. (2017). Mind Your Ps and Qs: The Interrelation between Period (P) and Mass-ratio (Q) Distributions of Binary Stars. *230*(2), 15. https://doi.org/10.3847/1538-4365/aa6fb6

Naab, T., & Ostriker, J. P. (2017). Theoretical Challenges in Galaxy Formation. *55*(1), 59–109. https://doi.org/10.1146/annurev-astro-081913-040019

Neuhäuser, R., Gießler, F., & Hambaryan, V. V. (2020). A nearby recent supernova that ejected the runaway star $\zeta$ Oph, the pulsar PSR B1706-16, and $^{60}$Fe found on Earth. *498*(1), 899–917. https://doi.org/10.1093/mnras/stz2629

Nomoto, K., Kobayashi, C., & Tominaga, N. (2013). Nucleosynthesis in Stars and the Chemical Enrichment of Galaxies. *51*(1), 457–509. https://doi.org/10.1146/annurev-astro-082812-140956

Offner, S. S. R., Moe, M., Kratter, K. M., Sadavoy, S. I., Jensen, E. L. N., & Tobin, J. J. (2023). The Origin and Evolution of Multiple Star Systems. In S. Inutsuka, Y. Aikawa, T. Muto, K. Tomida, & M. Tamura (Eds.), *Protostars and planets VII* (Vol. 534, p. 275). https://doi.org/10.48550/arXiv.2203.10066

Podsiadlowski, Ph., Joss, P. C., & Hsu, J. J. L. (1992). Presupernova Evolution in Massive

DRAFT

Interacting Binaries. *391*, 246. https://doi.org/10.1086/171341

Renzo, M., Zapartas, E., de Mink, S. E., Götberg, Y., Justham, S., Farmer, R. J., Izzard, R. G., Toonen, S., & Sana, H. (2019). Massive runaway and walkaway stars. A study of the kinematical imprints of the physical processes governing the evolution and explosion of their binary progenitors. *624*, A66. https://doi.org/10.1051/0004-6361/201833297

Repetto, S., Davies, M. B., & Sigurdsson, S. (2012). Investigating stellar-mass black hole kicks. *425*(4), 2799–2809. https://doi.org/10.1111/j.1365-2966.2012.21549.x

Repetto, S., Igoshev, A. P., & Nelemans, G. (2017). The Galactic distribution of X-ray binaries and its implications for compact object formation and natal kicks. *467*(1), 298–310. https://doi.org/10.1093/mnras/stx027

Repetto, S., & Nelemans, G. (2015). Constraining the formation of black holes in short-period black hole low-mass X-ray binaries. *453*(3), 3341–3355. https://doi.org/10.1093/mnras/stv1753

Röpke, F. K., & De Marco, O. (2023). Simulations of common-envelope evolution in binary stellar systems: physical models and numerical techniques. *Living Reviews in Computational Astrophysics*, *9*(1), 2. https://doi.org/10.1007/s41115-023-00017-x

Sana, H., de Mink, S. E., de Koter, A., Langer, N., Evans, C. J., & al., et. (2012). Binary Interaction Dominates the Evolution of Massive Stars. *Science*, *337*(6093), 444. https://doi.org/10.1126/science.1223344

Somerville, R. S., & Davé, R. (2015). Physical Models of Galaxy Formation in a Cosmological Framework. *53*, 51–113. https://doi.org/10.1146/annurev-astro-082812-140951

Zapartas, E., de Mink, S. E., Izzard, R. G., Yoon, S.-C., Badenes, C., Götberg, Y., de Koter, A., Neijssel, C. J., Renzo, M., Schootemeijer, A., & Shrotriya, T. S. (2017). Delay-time distribution of core-collapse supernovae with late events resulting from binary interaction. *601*, A29. https://doi.org/10.1051/0004-6361/201629685

DRAFT